10.8 Sample Programs to illustrate classes and objects

Program 1: Write a program to calculate area and circumference of a circle

```
class Circle:
        pi=3.14
def __init__(self,radius):
        self.radius=radius
def area(self):
        return Circle.pi*(self.radius**2)
def circumference(self):
        return 2*Circle.pi*self.radius
r=int(input("Enter Radius: "))
C=Circle(r)
print("The Area =",C.area())
print("The Circumference =", C.circumference())
```

Output:

```
Enter Radius: 5
The Area = 78.5
The Circumference = 31.400000000000002
```

Program 2: Write a program to accept a string and print the number of uppercase, lowercase, vowels, consonants and spaces in the given string

```
class String:
        def __init__(self):
                self.upper=0
                self.lower=0
                self.vowel=0
                self.consonant=0
                self.space=0
                self.string=""
        def getstr(self):
                self.string=str(input("Enter a String: "))
        def count (self):
                for ch in self.string:
                        if (ch.isupper()):
                                self.upper+=1
                        if (ch.islower()):
                                self.lower+=1
                        if (ch in ('AEIOUaeiou'):
                                self.vowel+=1
```


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